

# **Network UPS Tools Change Log**

NUT project community contributors

| REVISION HISTORY                   |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |
|------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| NUMBER                             | DATE       | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | NAME |
| 2.8.5.494 2.8.5.534-534+g9b8e1c06b | 2026-06-16 | Current release snapshot of Network UPS Tools (NUT).                                                                                                                                                                                                                                                                                                                                                                                                                              |      |
| 2.8.5                              | 2026-04-07 | Some changes to docs and recipes, especially for parallel builds; introduced a way to build with private shared libraries to save space. Updated NUT SEMVER definition some more, added a normalization and comparison script. Some rounds of code-hardening project. Added reference RPM/DEB recipes to test NUT iterations on OBS. Numerous driver updates, some new ones introduced.                                                                                           | JK   |
| 2.8.4                              | 2025-08-07 | Fixed a few regressions introduced by release v2.8.3. Some changes to docs and recipes, especially for parallel builds. Updated NUT SEMVER definition some more. Some rounds of code-hardening project. Numerous driver updates, some new ones introduced.                                                                                                                                                                                                                        | JK   |
| 2.8.3                              | 2025-04-07 | Some changes to docs and recipes, libupsclient API and functionality. Updated NUT SEMVER definition and added scripting around it. Groundwork for vendor-defined status and INSTCMD buzzwords like "ECO". Fixed some regressions and added improvements for certain new device series.                                                                                                                                                                                            | JK   |
| 2.8.2                              | 2024-04-01 | Some changes to docs and recipes, libnutscan API and functionality. Added nutconf (library and tool). Fixed some regressions and added improvements for certain new device series.                                                                                                                                                                                                                                                                                                | JK   |
| 2.8.1                              | 2023-10-31 | Some changes to API, docs and recipes, in particular to simplify local builds and tests (e.g. to help end-users check if current NUT codebase trunk has already fixed an issue they see with a packaged installation). Revived NUT for Windows effort, further improved other OS integrations. NUT became reference for "UPS management protocol", Informational RFC 9271. Documentation files refactored to ease maintenance. More drivers and new driver categories introduced. | JK   |

| REVISION HISTORY |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |
|------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| NUMBER           | DATE       | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | NAME |
| 2.8.0            | 2022-04-26 | Change of maintainer. Many changes to API, docs (both style and content), and recipes, with a stress on non-regression test-ability, run-time debug-ability, general codebase maintainability, as well as OS integrations (notably nut-driver-enumerator for systemd and SMF service instance maintenance). Added a lot in area of CI support and documented pre-requisite package lists for numerous platforms, and CI agent set-up. Added libusb-1.x support and many new driver categories (and drivers), and daisychain device connection support. Instant commands enhanced with TRACKING to enable protocol-based waiting for completion of a particular INSTCMD or SET operation. | JK   |
| 2.7.4            | 2016-03-09 | NUT variables namespace updated, in particular for outlet groups, alarms and thresholds, ATS devices, and battery.charger.status to supersede CHRG and DISCHRG flags published in ups.status readings. NUT network protocol extended with NUMBER type; some API changes.                                                                                                                                                                                                                                                                                                                                                                                                                 | AQ   |
| 2.7.3            | 2015-04-22 | Documentation revised, including some API changes. Added NUT DDL links. NUT variables namespace updated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | AQ   |
| 2.7.2            | 2014-04-17 | The nut-website project was offloaded into a separate repository. FreeDesktop HAL support was removed (obsoleted in GNOME consumer). Introduced nutdrv_atcl_usb driver.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | AQ   |
| 2.7.1            | 2013-11-19 | NUT source codebase migrated from SVN to Git (and from Debian infrastructure to GitHub source code hosting). jNut binding split into a separate project. Introduced libnutclient (C++ binding), al175, apcupsd-ups and nutdrv_qx drivers, Mozilla NSS support for simpler licensing than OpenSSL, and a newer apcsmart implementation. Documentation support enhanced with a spell checker, contents massively updated to reflect project changes.                                                                                                                                                                                                                                       | CL   |

| REVISION HISTORY |            |                                                                                                                                                                                            |      |
|------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| NUMBER           | DATE       | DESCRIPTION                                                                                                                                                                                | NAME |
| 2.6.5            | 2012-08-08 | New macosx-ups driver, new implementation of mge-shut driver. NUT variables namespace updated. Docs cleaned up and revised.                                                                | AQ   |
| 2.6.4            | 2012-05-31 | New NUT network protocol commands (LIST CLIENTS, LIST RANGE and NETVER), and socket protocol commands (ADDRANGE, DELRANGE). NUT variables namespace updated. Introduced nut-recorder tool. | AQ   |
| 2.6.3            | 2012-01-04 | No substantial changes to documentation.                                                                                                                                                   | AQ   |
| 2.6.2            | 2011-09-15 | Introduced nut-scanner tool and nut-ipmipsu driver, systemd support, and a new apcsmart implementation.                                                                                    | AQ   |
| 2.6.1            | 2011-06-01 | Introduced default.* and override.* optional settings in ups.conf, an ups.efficiency report, and outlet.0 special handling.                                                                | AQ   |
| 2.6.0            | 2011-01-14 | First release of AsciiDoc documentation for Network UPS Tools (NUT).                                                                                                                       | AQ   |

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| 2.5      | 2026-06-11 mbv06 <mbv06.dev@gmail.com> . . . . .                                         | 1        |
| 2.6      | 2026-06-11 Jim Klimov <jimklimov+nut@gmail.com> . . . . .                                | 1        |
| 2.7      | 2026-06-10 dependabot[bot] <49699333+dependabot[bot]@users.noreply.github.com> . . . . . | 2        |
| 2.8      | 2026-06-09 mbv06 <mbv06.dev@gmail.com> . . . . .                                         | 2        |
| 2.9      | 2026-06-08 mbv06 <mbv06.dev@gmail.com> . . . . .                                         | 2        |
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| 2.12     | 2026-06-05 Jim Klimov <jimklimov+nut@gmail.com> . . . . .                                | 2        |
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| 2.21     | 2026-05-25 Jim Klimov <jimklimov+nut@gmail.com> . . . . .                                | 4        |
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| 2.27     | 2026-05-18 Lukas Schmid <lukas.schmid@netcube.li> . . . . .                              | 5        |
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| 2.30     | 2026-05-15 Lukas Schmid <lukas.schmid@netcube.li> . . . . .                              | 6        |
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| 2.39 | 2026-05-09 Jim Klimov <jimklimov+nut@gmail.com>              | 8  |
| 2.40 | 2026-05-08 ostrich <570911+ostrich@users.noreply.github.com> | 8  |
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| 2.65 | 2026-04-14 Lukas Schmid <lukas.schmid@netcube.li>            | 17 |
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# 1 Introduction

The primary goal of the Network UPS Tools (NUT) project is to provide support for Power Devices, such as Uninterruptible Power Supplies, Power Distribution Units and Solar Controllers.

## 2 Very detailed Change Log

This document intends to detail the change log for relatively recent work (roughly since the source code was tracked in Git).

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### Note

This change log section represents git commits in range `v2.8.5..HEAD` (commits `0e051f9f6..9b8e1c06b`).

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### 2.1 2026-06-16 Jim Klimov <jimklimov+nut@gmail.com>

- docs/nut.dict: sort the dictionary after [PR #3447](#)
- NEWS.adoc: mention [PR #3448](#)
- NEWS.adoc: mention [PR #3456](#)
- docs/Makefile.am: fix ASPELL\_ENV\_LC\_ALL for platforms that care
- NEWS.adoc: refer to [PR #3485](#)

### 2.2 2026-06-15 Jim Klimov <jimklimov+nut@gmail.com>

- docs/ci-farm-do-setup.adoc, docs/nut.dict: add notes about ZFS and 4KB disk sectors

### 2.3 2026-06-14 mbv06 <mbv06.dev@gmail.com>

- NEWS.adoc, data/driver.list.in, drivers/nutdrv\_qx.c: nutdrv\_qx: Fix armac subdriver failing on 7-byte interrupt reads Certain UPS models (e.g. SVC VP-1250-LCD) return a 7-byte report instead of the standard 6 bytes. This relaxes the strict equality check in the interrupt read loop and increases the requested read size to 8 bytes to safely accommodate these devices without dropping trailing bytes or timing out.

### 2.4 2026-06-12 Jim Klimov <jimklimov+nut@gmail.com>

- Jenkinsfile-dynamatrix: update optional traceable toggles

### 2.5 2026-06-11 mbv06 <mbv06.dev@gmail.com>

- NEWS.adoc: docs: note Windows state pipe security update

### 2.6 2026-06-11 Jim Klimov <jimklimov+nut@gmail.com>

- drivers/richcomm\_usb.c, NEWS.adoc: bump DRIVER\_VERSION and document the improvements [[#1768](#), [#3477](#)]
-



## 2.7 2026-06-10 dependabot[bot] <49699333+dependabot[bot]@users.noreply.github.com>

- .github/workflows/01-make-dist.yml: build(deps): bump bluwy/substitute-string-action from 3 to 4 Bumps [bluwy/substitute-string-action](#) from 3 to 4. - [Release notes](#) - [Commits](#) ---- updated-dependencies: - dependency-name: bluwy/substitute-string-action dependency-version: 4 dependency-type: direct:production update-type: version-update:semver-major ----
- .github/workflows/01-make-dist-worker.yml: build(deps): bump actions/upload-artifact from 4 to 7 Bumps [actions/upload-artifact](#) from 4 to 7. \* [Release notes](#) \* [Commits](#) ---- updated-dependencies: \* dependency-name: actions/upload-artifact dependency-version: 7 dependency-type: direct:production update-type: version-update:semver-major ----

## 2.8 2026-06-09 mbv06 <mbv06.dev@gmail.com>

- docs/man/richcomm\_usb.txt, drivers/richcomm\_usb.c, tools/nut-scanner/scan\_usb.c: Fix richcomm USB matching options
- drivers/richcomm\_usb.c: drivers: preserve libusb context across USB reopens (richcomm only)

## 2.9 2026-06-08 mbv06 <mbv06.dev@gmail.com>

- drivers/dstate.c: drivers: fix Windows state pipe permissions

## 2.10 2026-06-08 Jim Klimov <jimklimov+nut@gmail.com>

- scripts/perl/UPS/Nut.pm: fix old trailing whitespaces

## 2.11 2026-06-06 Jim Klimov <jimklimov+nut@gmail.com>

- docs/Makefile.am: fix SOURCE\_CHECK\_UNICODE for out-of-tree builds

## 2.12 2026-06-05 Jim Klimov <jimklimov+nut@gmail.com>

- clients/Makefile.am, configure.ac: configure.ac: revise LIBSSL\_LIBS vs "-pthread" option (from NSS occasionally) for C++ builds [[#3453](#)] Address error: argument unused during compilation: *-pthread*
- Makefile.am, SECURITY.md, docs/Makefile.am: SECURITY.md: introduce a policy that can be seen on GitHub Describe the current situation with support of NUT versions and the parties involved, and how to responsibly disclose vulnerabilities.

## 2.13 2026-06-03 Jim Klimov <jimklimov+nut@gmail.com>

- .github/workflows/01-make-dist-worker.yml, .github/workflows/01-make-dist.yml, .github/workflows/05-codeql.yml, .github/workflows/PyNUTClient.yml: .github/workflows/\*.yml: bump actions versions Originally proposed by dependabot, but we cannot put its Markdown commit comments into ChangeLog.pdf, so reposting.
  - .github/dependabot.yml: introduce scanning for obsolete GHA versions
  - .github/workflows/01-make-dist.yml: update comments about the Checks message [[#1400](#)]
  - .github/workflows/01-make-dist.yml: when dispatching the workflow for a branch iteration, use its name (ref) not commit [[#1400](#)]
  - .github/workflows/01-make-dist.yml: delete old artifacts OTHER than the one just created [[#1400](#)] Before the recent refactoring, the logic was to first build the tarball, then if it succeeded - clean up the storage from obsolete ones, and quickly publish the new one. Now that a separate workflow does the uploads, we can at best ensure deletion of obsolete tarballs after hopefully having the new one up there (had enough space and all that). Also take care to not delete the newly made artifact ;)
  - NEWS.adoc: update for FreeBSD and Jenkins bumps
-

- NEWS.adoc: Rearranged the GHA job to prepare Dist and Docs Tarballs [#1400, #3460]
- .github/workflows/01-make-dist.yml: launch "GHA-01: Tarballs Build Worker" definition from the target branch (if a PR) [#1400]
- configure.ac: pass -Wno-unused-command-line-argument also to CLANG LDFLAGS
- .github/workflows/01-make-dist.yml: DRY: define artifact\_name once [#1400]

#### 2.14 2026-06-02 Jim Klimov <jimklimov+nut@gmail.com>

- .github/workflows/01-make-dist-worker.yml, .github/workflows/01-make-dist.yml: .github/workflows/01-make-dist{,-worker}.yml: retell the story of two workflows [#1400]
- .github/workflows/01-make-dist.yml: reduce the permissions needed for artifact manipulation [#1400]

#### 2.15 2026-06-01 Jim Klimov <jimklimov+nut@gmail.com>

- Jenkinsfile-dynamatrix: use new countStagesPerNode feature from the library in notifyHandlerAlert
- Jenkinsfile-dynamatrix: introduce dynacfgPipeline.notifyHandlerAlert (to report looping RESTART dynamatrix cells)

#### 2.16 2026-05-31 Jim Klimov <jimklimov+nut@gmail.com>

- configure.ac: ignore recent clang warning about unused-command-line-argument

#### 2.17 2026-05-31 Doug Nazar <nazard@nazar.ca>

- common/common.c, drivers/upsdrvquery.c, include/common.h, server/sstate.c: Use non-blocking connect() for AF\_UNIX sockets to ensure we don't stall If a driver has become unresponsive, connect() will block after the listen() backlog has been exhausted, causing the main upsd to also become unresponsive.

#### 2.18 2026-05-28 Jim Klimov <jimklimov+nut@gmail.com>

- tools/check-source-nonascii.pl: fix syntax for hash length
- ci\_build.sh: fix normalize\_path\_perl() syntax for hash length; avoid newlines

#### 2.19 2026-05-27 Jim Klimov <jimklimov+nut@gmail.com>

- docs/config-prereqs.txt: document JDK21+ as the requirement for NUT CI build farm agents since spring 2026
  - ci\_build.sh: normalize\_path() before reporting it at start of actual work about BUILD\_TYPE
  - ci\_build.sh: configure\_nut(): report the PATH value ultimately used (maybe normalized)
  - ci\_build.sh: propose shell vs perl (faster?) implementations of normalize\_path()
  - drivers/bicker\_ser.c, drivers/nut-libfreeipmi.c, drivers/phoenixcontact\_modbus.c, drivers/socomec\_jbus.c, drivers/usbhid-ups.c: drivers/\*.c: replace Unicode in comments with ASCII equivalents
  - drivers/socomec\_jbus.c: clean up trailing whitespace
  - docs/documentation.txt, docs/man/nut.txt: replace Unicode long dashes with ascii --
  - drivers/libwinhid.h: replace Unicode long dashes with ascii -- and arrow with ==>
-

- docs/developers.txt, docs/nut.dict: docs/developers.txt: add a chapter on ASCII vs Unicode
- Makefile.am, configure.ac, docs/Makefile.am, tools/Makefile.am, tools/check-source-nonascii.pl: configure.ac, Makefile.am, docs/Makefile.am, tools/check-source-nonascii.pl: add unicode-in-sources detection to spellcheck On some platforms, at least the `file` type parsing fails for text with some but not other wide characters (so it is "data" and not some sort of "text"). This test was added for C sources as well, but keep in mind that the final authority is the actually used compilers - which are free to complain or not.
- .github/pull\_request\_template.md: update with a section on Unicode characters
- configure.ac: report the tested compiler location just before `AC_PROG_CC` etc. which might fail
- configure.ac: `unset CCACHE_DIR` if provided but empty Notably, `ccache-3.7.12` in FreeBSD 15 breaks CI build with that complaint.
- drivers/ragtech.c: pre-declare variables for portability [#3447]
- drivers/ragtech.c: replace Unicode long dashes with `acsii` -- [#3447]
- docs/man/ragtech.txt: replace Unicode long dashes with `acsii` -- [#3447]

## 2.20 2026-05-26 Jim Klimov <jimklimov+nut@gmail.com>

- docs/config-prereqs.txt, docs/nut.dict: revise with FreeBSD 15 lessons learned
- configure.ac: if developer enables configure debug, only wipe `DOCTESTDIR` after confirming we can build docs we want
- drivers/ragtech.c: fix compiler warnings about `%u` vs. conditionally picked numbers [#3447] ... which are signed int by default

## 2.21 2026-05-25 Jim Klimov <jimklimov+nut@gmail.com>

- tests/NIT/nit.sh: `check_NIT_certs_NSS()`: fix the check for `*.txt` and `*.db` file patterns [#3331, #1711]
- tests/NIT/nit.sh: wrap long openssl lines; fix a missed validity-time spec [#1711]
- tests/NIT/nit.sh: `check_NIT_certs_NSS()`: verify old run-time toolkit vs. newer (cached) DB files [#3331, #1711] Auto-repair the trusted NSS DB set(s) of files if we can. Primarily relevant for build agents where same cache is used by NSS toolkits a decade apart.
- docs/nut.dict: add opcodes
- configure.ac: add `-Wno-thread-safety-negative` to clang flags (seen on FreeBSD builds)

## 2.22 2026-05-24 Pedro Cunha <pedroagracio+nut@gmail.com>

- drivers/libusb1.c: use explicit context; drop per-close `libusb_exit()` [#598] The original code called `libusb_init(NULL)` on every `nut_libusb_open()` and `libusb_exit(NULL)` on every `nut_libusb_close()`, bumping and decrementing the default context's refcount. That violates the documented contract for `libusb_exit`, which "should be called after closing all open devices and before your application terminates", and on certain firmware it wedges indefinitely: when `libusb_exit` hits the `1→0` transition it tears the context down, which waits on `libusb`'s internal sync primitives for outstanding URBs to drain. URBs orphaned by `libusb_reset_device` or by an unexpected device disconnect never drain, so the wait never returns. The deadlock is reachable from any reconnect path. It was first recognized by `4f84b7f92` ("don't `libusb_exit()` when closing a previously opened device"), which dropped the call in `nut_libusb_open()`'s rematch loop. The fallback reconnect path in `qx_command` and the rest of the driver lifecycle still hit it. With the companion change in `nutdrv_qx.c` (`a4813f868`) that escalates persistent `LIBUSB_ERROR_OVERFLOW` to `libusb_reset_device`, the deadlock window opens on every escalation and was hit reproducibly on the `0665:5161` Cypress USB-serial bridge family (Salicru SPS, Ippon, ViewPower, Voltronic Power UPSes; see [#598, #993, #2453]). Switch to an explicit `libusb` context owned end-to-end: - Initialize once on the first `nut_libusb_open()` and register a matching one-shot `atexit` handler. - Remove `libusb_exit()` from `nut_libusb_close()`; the close path now only calls `libusb_close()`, which is non-blocking. - Replace `libusb_get_device_list(NULL, ...)` with the explicit context so the default context is never touched.

- Bound the `atexit libusb_exit()` with a SIGALRM-based 2s timeout (POSIX) so a deadlock-prone teardown at shutdown can't wedge supervisor stop sequences (e.g. `systemctl stop`). Sub-millisecond on a healthy context. Bump `USB_DRIVER_VERSION` from 0.53 to 0.54. Validated on a Salicru SPS 1500 ONE BL (0665:5161) by authorize-toggle stress reproduction (8 cycles, clean reconnect each time, same driver PID throughout) and by a natural OVERFLOW event captured in soak (counter-gated logic at 1/3, next poll succeeded, no escalation; driver PID stable for 22h\+).

## 2.23 2026-05-22 Jim Klimov <jimklimov+nut@gmail.com>

- `lib/libupsclient-config.in`: this should consider `LIBSSL_LDFLAGS_RPATH` [#3331]
- `lib/libnutclient.pc.in`: this now Requires: `@LIBSSL_REQUIRES@` [#3331]

## 2.24 2026-05-20 Pedro Cunha <pedroagracio+nut@gmail.com>

- `drivers/nutdrv_qx.c`: `qx_command()`: reset USB device after persistent `LIBUSB_ERROR_OVERFLOW` [#598] `LIBUSB_ERROR_OVERFLOW` has been handled as a benign, retry-on-next-poll condition (grouped with `LIBUSB_ERROR_TIMEOUT` and the default case) ever since the original blazer import. For most devices a one-off oversized interrupt-IN frame is indeed transient. But some Cypress USB-serial bridge firmware (VID:PID 0665:5161; Salicru SPS, Ippon, ViewPower and various Voltronic Power UPSes) wedges the endpoint once it overruns: every subsequent interrupt read returns `OVERFLOW` and the driver spins in the stale-data loop until an external USB-level reset. This is the chronic hang reported in [issue #598](#). Distinguish the two cases with a small consecutive-overflow counter: the first `QX_USB_OVERFLOW_RESET_TRIES-1` overflows are retried on the next poll (so genuine transients cost nothing but a skipped cycle), and a sustained run escalates to `usb_reset()` \+ reconnect, reusing the recovery path already used for `PIPE/ETIME/IO` errors. Any clean read resets the counter. Captured on a Salicru SPS 1500 ONE BL: an isolated overflow recovers on the next poll, while a sustained lockup (dozens of consecutive overflows within seconds) is cleared only by the device reset, matching the behaviour of an external `usb_resetter --reset-device`.

## 2.25 2026-05-20 Jim Klimov <jimklimov+nut@gmail.com>

- `clients/upsstats.c`: `ups_connect()`: avoid potential NULL dereference in debug/failure printouts

## 2.26 2026-05-19 Jim Klimov <jimklimov+nut@gmail.com>

- `tests/NIT/Makefile.am`: `check-NIT-devel`: minimize `nut_version.h` regeneration attempts [#1711]
- `tests/NIT/nit.sh`: fix passing server cert trust to NSS DB [#3331, #1711]
- `tests/NIT/nit.sh`: fix passing server cert usage to NSS DB [#3331, #1711]
- `tests/NIT/nit.sh`: fix passing root CA cert trust to NSS DB [#3331, #1711]

## 2.27 2026-05-18 Lukas Schmid <lukas.schmid@netcube.li>

- `drivers/apc_common.c`, `drivers/apc_common.h`: add Axel Gembe copyright to `apc_common` as the code is at the very least inspired by the original code from `apc-modbus`

## 2.28 2026-05-18 Jim Klimov <jimklimov+nut@gmail.com>

- `drivers/apcmicrolink.c`: drop pragma to ignore a warning unknown to old clang [#3406]
-

**2.29 2026-05-17 juslex <66561713+juslex@users.noreply.github.com>**

- NEWS.adoc, data/driver.list.in, docs/man/Makefile.am, docs/man/ragtech.txt, docs/nut.dict, drivers/Makefile.am, drivers/ragtech.c, include/wincompat.h: ragtech: new driver for Ragtech UPSes (USB CDC-ACM, family 10) Targets Brazilian-built Ragtech "Easy Pro" / NEP / TORO / INNERGIE / OneUP devices that present themselves as USB CDC-ACM (VID 0x04D8, PID 0x000A, Microchip PIC firmware). Validated end-to-end against an Easy 2000 TI (reg 0x9A model id = 16) read out of a working OneUP Nitro 2000. Protocol — three opcodes observed in OEM traffic: 0x01 ADDR\_HI ADDR\_LO VALUE CKSUM write byte 0x02 ADDR\_HI ADDR\_LO MASK CKSUM AND mask (atomic bit-clear) 0x04 ADDR\_HI ADDR\_LO COUNT CKSUM read range CKSUM = (ADDR\_HI \+ ADDR\_LO + VALUE) & 0xFF The CDC-ACM channel ignores baud at the wire but DTR/RTS are interpreted by some Ragtech families as a remote shutdown signal — the driver forces both low after open and does NOT call ser\_set\_speed() to avoid the tcsetattr() DTR pulse that some Linux tty drivers perform. The full 30-byte main range (0x80..0x9D) plus V\_IOUTCALIB (0xF3) and the oscillator calibration pair (0x202/0x203) are read; output frequency is interpolated per devices.xml formula rather than hard-coded. Twenty models from the family-10 device table populate ups.model, ups.realpower.nominal, output.voltage scaling, output.current scaling and battery voltage scaling. input.voltage.nominal switches between 115V and 220V dynamically based on the measured input. Instcmds implemented: shutdown.stayoff — aa 02 00 80 fe (clear AUTOSTART) \+ aa 01 00 98

A. Byte-for-byte identical to the OEM supsvc sequence. Validated: cuts output and stays off until manual power-on, in both AC and battery modes. shutdown.return — falls back to shutdown.stayoff with a warning. The atomic OR opcode that would set F\_AUTOSTART has not been captured yet; read-modify-write with 0x01 sends the bytes but does not trip the firmware's shutdown state machine. shutdown.stop — aa 01 00 98 00 (write V\_SHUTDOWNTIMER = 0). Confirmed to abort an armed countdown. test.battery.start.deep — bit writes on regs 0x90 / 0x95 (fullDischarge in devices.xml). Flagged as unverified for the same reason as shutdown.return. Read primitive uses ser\_get\_buf() in a manual loop rather than ser\_get\_buf\_len() (which discards partial reads on timeout — CDC fragments replies into 1-8 byte chunks). ser\_flush\_io() is used before each TX to clear the kernel CDC buffer (ser\_flush\_in() only does a select\+read loop that misses queued bytes under O\_NONBLOCK). Reverse engineering primarily from the OEM devices.xml table for register layout / scaling / flags / actions, and from trace of the OEM supsvc binary capturing both read polls and the LED/shutdown write sequences. Cross-checked against UPS\_ESP32\_tinySrv ([https://github.com/antunesls/UPS\\_ESP32\\_tinySrv](https://github.com/antunesls/UPS_ESP32_tinySrv)).

**2.30 2026-05-15 Lukas Schmid <lukas.schmid@netcube.li>**

- docs/man/apcmicrolink.txt, docs/nut.dict, drivers/apcmicrolink.c: update docs and spellcheck dictionary

**2.31 2026-05-14 Lukas Schmid <lukas.schmid@netcube.li>**

- drivers/Makefile.am: add missing apc\_common.h to distributed headers
- drivers/apcmicrolink-maps.c, drivers/apcmicrolink-maps.h, drivers/apcmicrolink.c: extract descriptor helpers and fix map includes

**2.32 2026-05-13 Lukas Schmid <lukas.schmid@netcube.li>**

- drivers/apcmicrolink.c, drivers/apcmicrolink.h: fix switch warnings and replace flag bit-fields

**2.33 2026-05-13 Jim Klimov <jimklimov+nut@gmail.com>**

- NEWS.adoc, drivers/nutdrv\_qx\_megatec.c: drivers/nutdrv\_qx\_megatec.c: add QX\_FLAG\_ABSENT to "ups.firmware" so devices with all-spaces in that field can still be recognized [#3436] Co-authored-by: Jim Klimov <jimklimov+nut@gmail.com> Co-authored-by: Siravijbb <siravijbb@gmail.com>

**2.34 2026-05-12 Lukas Schmid <lukas.schmid@netcube.li>**

- drivers/apc\_modbus.c: remove unused function
-

### 2.35 2026-05-12 Jim Klimov <jimklimov+nut@gmail.com>

- common/common.c: cosmetic fix
- docs/Makefile.am: add ASPELL\_NUT\_COMMON\_ARGS to preclude mangling of nut.dict on some platforms when checking interactively

### 2.36 2026-05-11 Lukas Schmid <lukas.schmid@netcube.li>

- drivers/apcmicrolink.c: fix commands and writes sometimes not working and allow configuration of command source
- drivers/Makefile.am, drivers/apc\_common.c, drivers/apc\_common.h, drivers/apc\_modbus.c, drivers/apc\_modbus.h, drivers/apcmicrolink-maps.c, drivers/apcmicrolink-maps.h, drivers/apcmicrolink.c: Variable cleanup and commons creation between apc-modbus and microlink

### 2.37 2026-05-11 Jim Klimov <jimklimov+nut@gmail.com>

- server/upsd.c: update\_sysmaxconn(): default the sysmaxconn value to "MAX - RESERVE\_FD\_COUNT\_UPSD" on platforms where it might matter [#3365]
  - server/upsd.c: update\_sysmaxconn(): check that detected sysmaxconn\_hard is big enough to be useful considering the RESERVE\_FD\_COUNT\_UPSD set aside AND a few actual connections [#3365]
  - server/upsd.c: update\_sysmaxconn(): revise warning message about hard limit [#3365]
  - server/upsd.c: be sure to re-detect sysmaxconn after (re-)loading config [#3365]
  - NEWS.adoc, UPGRADING.adoc, docs/nut.dict, server/upsd.c, server/upsd.h: server/upsd.{c,h}, NEWS.adoc, UPGRADING.adoc: consider getrlimit() permissions vs. MAXCONN request [#3365]
  - NEWS.adoc, server/conf.c, server/upsd.c, server/upsd.h: server/conf.c, server/upsd.{c,h}, NEWS.adoc: if we hit "Too many open files" during configuration reload, close oldest client connection and retry [#3365]
  - server/upsd.h: update © heading and whitespace styling
  - docs/nut.dict: update for dllldd [#3420]
  - NEWS.adoc: update about scripts/Windows/dllldd.sh helper script refinements [#3420]
  - docs/man/upscli\_add\_host\_cert.txt: update that this call works also with OpenSSL (since NUT v2.8.6) [#3331]
  - docs/download.txt, docs/nut.dict: Upload NUT-for-Windows-x86\_64-RELEASE-2.8.5-1-fixNSS.7z with NSS dependency libraries missing in original NUT v2.8.5 release [#3420]
  - appveyor.yml: allow ln to create hard links, and tell 7-zip to store them as such, not as independent files (subject to 7zip format du jour)
  - scripts/Windows/dllldd.sh: Revise use of TEMPFILE\_REC to avoid grep if it is empty [#3420]
  - Makefile.am: install-win-bundle-thirdparty: revise progress message markup for easier troubleshooting [#3420]
  - scripts/Windows/dllldd.sh: dllldd\_with\_strings(): filter away OUT\_TOOLS hits if known [#3420]
  - scripts/Windows/dllldd.sh: dllldd\_with\_strings(): quick bail-out with TEMPFILE\_REC [#3420]
  - scripts/Windows/dllldd.sh: revise cleanup of TEMPFILE\_REC; create it earlier in dllldddir() [#3420]
  - scripts/Windows/dllldd.sh: dllldd(): use TEMPFILE\_REC if defined by caller [#3420]
  - scripts/Windows/dllldd.sh: dllldd\_with\_tools(), dllldd(): optimize looped iteration of search paths we prepared [#3420] Avoid calls to tr ' : ' '\n' that can be done once.
  - scripts/Windows/dllldd.sh: relocate TEMPFILE\_REC declaration to before first use-mention [#3420]
-



- scripts/Windows/dllldd.sh: filter\_away\_system\_DLLs(): add some more OS-provided common hits to exclusion list [#3420]
- scripts/Windows/dllldd.sh: filter\_away\_system\_DLLs(): add DWrite.dll [#3420] Some OS releases have Microsoft DirectWrite for high-quality text rendering, anti-aliasing, and font management... not something we really care about in NUT, and part of OS footprint we do not drill into.
- scripts/Windows/dllldd.sh: dllldd\_with\_tools(), dllldd(): optimize SEARCH\_INPUT\_PATH discovery [#3420] Instead of a loop with dirname calls (spawning those processes takes a few seconds in MSYS2 on Windows), use one sed process.
- scripts/Windows/dllldd.sh: dllldd\_with\_strings(): ignore also "lib%s.dll" hits [#3420]
- scripts/Windows/dllldd.sh: accept also outputs from "objdump" variants that return "NEEDED libname" [#3420]
- scripts/Windows/dllldd.sh: when given a CLI argument, check if it is a directory (to run dllldddir) before diving in with dlllddrec [#3420]
- scripts/Windows/dllldd.sh: accept also outputs from "ldd" variants that only return path (no parenthesis with hex) [#3420]
- scripts/Windows/dllldd.sh: allow to parameterize DLLEXTR\_REGEX [#3420] At least, allow to test/iterate the script with non-Windows builds too.
- scripts/Windows/dllldd.sh: dllldd\_with\_strings(): filter away some bogus strings outputs and findings [#3420]
- scripts/Windows/dllldd.sh: more filter\_away\_system\_DLLs() [#3420]

### 2.38 2026-05-10 Jim Klimov <jimklimov+nut@gmail.com>

- scripts/Windows/dllldd.sh: dllldd\_with\_tools(): define and clean-up TEMPFILE\_REC in the outer-most call if we stack them [#3420] This should help directory crawlers etc. ignore paths already walked by some previous recursion in this scripted execution.
- scripts/Windows/dllldd.sh: dllldd\_with\_tools(): do not bail out if we found some but not all DLLs via objdump; feed only the names we did not resolve to ldd [#3420]
- scripts/Windows/dllldd.sh: consider also SEARCH\_INPUT\_PATH based on where the input EXE/DLL files are [#3420] This allows to find libnutprivate\*.dll or libupsclient\*.dll files we built and installed to EXE bindir, which were otherwise ignored by earlier path discovery tools and/or logic.
- scripts/Windows/dllldd.sh: filter\_away\_NUT\_DLLs() when we query both dllldd\_with\_tools() and dllldd\_with\_strings() [#3420]
- scripts/Windows/dllldd.sh: refactor with discover\_COMPILER\_PATHS() to not do this in a loop [#3420]

### 2.39 2026-05-09 Jim Klimov <jimklimov+nut@gmail.com>

- docs/nut.dict: Update nut.dict [#3430]

### 2.40 2026-05-08 ostrich <570911+ostrich@users.noreply.github.com>

- scripts/python/app/NUT-Monitor-py3qt5.in, scripts/python/app/NUT-Monitor-py3qt6.in: Fix NUT-Monitor tray tooltip text

### 2.41 2026-05-08 Jim Klimov <jimklimov+nut@gmail.com>

- NEWS.adoc: update about NUT-Monitor tooltips [#3430]

### 2.42 2026-05-07 Jim Klimov <jimklimov+nut@gmail.com>

- clients/upsmon.c, NEWS.adoc: do not abort if we did not succeed upscli\_init() but did not require SSL in the first place [#3420]
  - scripts/Windows/dllldd.sh: refactor to find DLLs with strings too; track filenames we already investigated [#3420] DRY the loop and avoid cycles.
-

**2.43 2026-05-06 Jim Klimov <jimklimov+nut@gmail.com>**

- m4/nut\_check\_libnetsnmp.m4: revise with AC\_CACHE\_VAL() for each option to be remembered [#3108]
- m4/nut\_check\_libavahi.m4: revise with AC\_CACHE\_VAL() for each option to be remembered [#3108]
- ci\_build.sh, scripts/Windows/build-mingw-nut.sh: get\_CI\_CACHE\_NUT\_HASHDIR\_CFG\_OPT(): consider also MAKE and (CONFIG\_)SHELL as circumstances that impact the configuration hash [#3108]

**2.44 2026-05-05 Jim Klimov <jimklimov+nut@gmail.com>**

- m4/nut\_check\_libfreeipmi.m4: revise with AC\_CACHE\_VAL() for each option to be remembered [#3108]
- m4/nut\_check\_libgd.m4: revise with AC\_CACHE\_VAL() for each option to be remembered [#3108]
- m4/nut\_check\_libglib.m4, configure.ac: revise with AC\_CACHE\_VAL() for each option to be remembered [#3108]
- m4/nut\_check\_libgpod.m4: revise with AC\_CACHE\_VAL() for each option to be remembered [#3108]
- m4/nut\_check\_libmodbus.m4: revise with AC\_CACHE\_VAL() for each option to be remembered [#3108]
- m4/nut\_check\_libneon.m4: revise with AC\_CACHE\_VAL() for each option to be remembered [#3108]
- m4/nut\_check\_libltdl.m4: do not drag around nut\_have\_libltdl in cached section, use nut\_cv\_have\_libltdl right away [#3108]
- m4/nut\_check\_libopenssl.m4, m4/nut\_check\_libnss.m4, configure.ac: revise with AC\_CACHE\_VAL() for each option to be remembered [#3108]
- m4/nut\_check\_libpowerman.m4: revise with AC\_CACHE\_VAL() for each option to be remembered [#3108]
- m4/nut\_check\_libusb.m4: revise with AC\_CACHE\_VAL() for each option to be remembered [#3108]
- m4/nut\_check\_os.m4: fix comments starting a line with a hash to use "dnl" first

**2.45 2026-05-04 Jim Klimov <jimklimov+nut@gmail.com>**

- m4/nut\_check\_libregex.m4: revise with AC\_CACHE\_VAL() for each option to be remembered [#3108]
  - m4/nut\_check\_libsystemd.m4: revise with AC\_CACHE\_VAL() for each option to be remembered [#3108]
  - m4/nut\_check\_libwrap.m4: revise with AC\_CACHE\_VAL() for each option to be remembered [#3108]
  - m4/nut\_check\_pkgconfig.m4: revise with AC\_CACHE\_VAL() for each option to be remembered [#3108]
  - m4/nut\_check\_os.m4: set OS\_NAME and PKG\_TARGET after passing AC\_CACHE\_CHECK [#3108]
  - m4/nut\_check\_libltdl.m4: do not reset nut\_have\_libltdl=yes when checking for header from cached data [#3108]
  - m4/nut\_check\_libltdl.m4: add a summary print-out for cached re-runs [#3108]
  - common/common.c: upsnotify(): fix "unused" warnings in different build combos
  - ci\_build.sh, ci\_build.adoc: when we DO\_USE\_NUTCI\_CACHE\_DEBUG, stash also the help text from configure script [#3108]
  - m4/nut\_check\_pkgconfig.m4: fix indentations, use AS\_IF and AS\_CASE more [#3108]
  - m4/nut\_check\_libwrap.m4: fix indentations, use AS\_IF and AS\_CASE more [#3108]
  - m4/nut\_check\_libusb.m4: fix indentations, use AS\_IF and AS\_CASE more [#3108]
  - m4/nut\_check\_libsystemd.m4: fix indentations, use AS\_IF and AS\_CASE more [#3108]
  - m4/nut\_check\_libregex.m4: fix indentations, use AS\_IF and AS\_CASE more [#3108]
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- m4/nut\_check\_libpowerman.m4: fix indentations, use AS\_IF and AS\_CASE more [#3108]
- m4/nut\_check\_libopenssl.m4, m4/nut\_check\_libpowerman.m4: m4/nut\_check\_libopenssl.m4: fix indentations, use AS\_IF and AS\_CASE more [#3108]
- m4/nut\_check\_libnss.m4: fix indentations, use AS\_IF and AS\_CASE more [#3108]
- m4/nut\_check\_libnetsnmp.m4: fix indentations, use AS\_IF and AS\_CASE more [#3108]
- m4/nut\_check\_libneon.m4: fix indentations, use AS\_IF and AS\_CASE more [#3108]
- m4/nut\_check\_libmodbus.m4: fix indentations, use AS\_IF and AS\_CASE more [#3108]
- m4/nut\_check\_libgpod.m4: fix indentations, use AS\_IF and AS\_CASE more [#3108]
- m4/nut\_check\_libglib.m4: fix indentations, use AS\_IF and AS\_CASE more [#3108]
- m4/nut\_check\_libgd.m4: fix indentations, use AS\_IF and AS\_CASE more [#3108]
- m4/nut\_check\_libfreeipmi.m4: fix indentations, use AS\_IF [#3108]
- m4/nut\_check\_libavahi.m4: fix indentations, use AS\_IF [#3108]
- m4/nut\_check\_cppcheck.m4: fix indentations, use AS\_IF and AC\_CACHE\_VAL [#3108]
- m4/nut\_check\_bool.m4: fix indentations [#3108]
- m4/nut\_check\_aspell.m4: fix indentations, use AS\_IF [#3108]
- m4/nut\_check\_asciidoc.m4: fix indentations, use AS\_IF [#3108]
- m4/nut\_check\_libltdl.m4: fix indentations, use AS\_IF for outer check [#3108]
- m4/nut\_check\_os.m4: fix typo in heading
- m4/nut\_check\_libltdl.m4: revise with AC\_CACHE\_VAL() for each option to be remembered [#3108]
- Makefile.am: when we touch -d "-10 seconds", hide the errors (such request is known to be not too portable)
- ci\_build.sh, scripts/Windows/build-mingw-nut.sh: facilitate re-runs with changed .m4/.am/.ac sources and enabled autoconf cache: handle precious CCACHE\_args/vars [#3108]
- ci\_build.adoc: clarify some points about caching [#3108]

#### 2.46 2026-05-02 Jim Klimov <jimklimov+nut@gmail.com>

- ci\_build.sh, ci\_build.adoc: introduce DO\_USE\_AUTOCONF\_CACHE\_DEBUG=each mode [#3108]

#### 2.47 2026-05-01 Jim Klimov <jimklimov+nut@gmail.com>

- NEWS.adoc: move entry for #3302 into v2.8.6 release

#### 2.48 2026-04-30 Jim Klimov <jimklimov+nut@gmail.com>

- docs/config-prereqs.txt: do not pull valgrind into default Debian/Ubuntu footprint, it is not ubiquitous Revise other suggestions a bit too

## 2.49 2026-04-29 Jim Klimov <jimklimov+nut@gmail.com>

- tests/cpptest-client.cpp: revise reporting of UPSCLI\_SSL\_CAPS\* [#3331, #1711]
- clients/nutclient.cpp, clients/nutclient.h, clients/upsclient.c, clients/upsclient.h, docs/man/upscli\_ssl\_caps.txt: upsclient, nut-client, docs: introduce UPSCLI\_SSL\_CAPS\_CERTHOST\* [#3331]
- tests/NIT/nit.sh: consider if we can not do CERTHOST validation [#3331]
- clients/upsclient.c: report if we can not do CERTHOST(name) validation [#3331]
- Makefile.am: .libs-dev-PATH: fix typo

## 2.50 2026-04-28 Jim Klimov <jimklimov+nut@gmail.com>

- clients/Makefile.am, common/Makefile.am, scripts/obs/debian.rules, scripts/obs/nut.spec, tools/nut-scanner/Makefile.am: **/Makefile.am, scripts/obs/**: bump libnutclient ABI level [#1290, #3331]
- clients/nutclient.cpp: drop obsolete FIXME comments [#3331]
- clients/nutclient.cpp: with openssl builds, do not require non-trivial certhost\_name IF we do not have methods to check it (openssl too old) [#3331]
- tests/NIT/nit.sh: refactor by use of check\_NIT\_certs\_NSS() [#1711]
- tests/NIT/nit.sh: introduce check\_NIT\_certs\_NSS() [#1711]
- tests/cpptest-client.cpp: enable testing for CERTHOST with OpenSSL builds, update argument order for NSS builds [#3331, #1711]
- clients/Makefile.am, clients/nutclient.cpp, clients/nutclient.h, docs/man/libnutclient\_tcp.txt: clients/nutclient.{cpp,h}, docs, Makefile.am: fix SSLConfig\_NSS related argument order to place certhost after certident, to match logical proximity and OpenSSL equivalents; reset ABI age (not backwards compatible to recent v2.8.5 release) [#3331]
- tests/NIT/nit.sh: refactor by use of check\_NIT\_certs\_NSS() [#1711, #3390, #3108]
- tests/NIT/nit.sh: introduce check\_NIT\_certs\_NSS() [#1711]
- tests/NIT/nit.sh: revise sanity-check, not all NSS versions had a \*.txt file [#3331, #1711]
- tests/NIT/nit.sh: consider WITH\_SSL\_CLIENT\_CERTIDENT=none to (re)set certificate nickname verification [#3331, #1711]
- tests/NIT/nit.sh: reword SSL support messages [#1711]
- tests/NIT/nit.sh: revise sanity-check, not all NSS versions had a \*.txt file [#3331, #1711]

## 2.51 2026-04-27 Jim Klimov <jimklimov@gmail.com>

- clients/nutclient.cpp: fix NULL⇒nullptr [#3331]
  - tests/NIT/nit.sh: isTestablePython(): fix detection of interpreter shebang via /usr/bin/env [#1711]
  - tests/NIT/nit.sh: isTestablePython(): fix detection of interpreter shebang via /usr/bin/env [#1711]
  - tests/NIT/nit.sh: refactor and improve portability with somehash\_filter()/somehash\_files() method detection [#3390]
  - ci\_build.sh: refactor and improve portability with somehash\_filter()/somehash\_files() method detection [#3108]
-

## 2.52 2026-04-27 Jim Klimov <jimklimov+nut@gmail.com>

- .circleci/config.yml: allow various cache contents for same branch/job iterations [#3390]
- tests/NIT/nit.sh: introduce check\_NIT\_certs() to ensure availability of all expected certs for NSS or OpenSSL scenarios [#3390]
- Makefile.am: .libs-dev-PATH: adjust for MacOS [#1711]
- tests/cpptest-client.cpp: report also NUT\_PORT [#1711, #3331]
- tests/cpptest-client.cpp: fix reporting of incoming envvars [#1711, #3331]
- clients/upsclient.c, clients/nutclient.cpp: accept systems where openssl does not define X509\_V\_ERR\_HOSTNAME\_MISMATCH [#3331]
- clients/upsclient.c, clients/nutclient.cpp: use pragmas to hide unreachable code warnings when built against openssl without X509 checking methods we use [#3331]
- ci\_build.sh, tests/NIT/nit.sh: introduce DO\_USE\_NUTCI\_CACHE\_DEBUG to stash optional items not needed in production use of the cache, but rather to track its regressions on re-runs etc. [#3108]
- Makefile.am: use a more portable date spec in touch for start of Epoch
- configure.ac: properly use AC\_CACHE\_VAL for BSDKVMPROCLIBS [#3108]
- m4/nut\_check\_libltdl.m4: re-detect ltdl.h details if we needed to retry, to not lose track [#3108]
- tests/NIT/nit.sh: fix NUT\_CAPATH for PERL SSL tests on Windows (MSYS2) yet some more [#1711]
- tests/NIT/nit.sh: fix NUT\_CAPATH for PERL SSL tests on Windows (MSYS2) with forward slashes too [#1711]
- tests/NIT/nit.sh: fix NUT\_CAPATH for PERL SSL tests on Windows (MSYS2) with forward slashes too [#1711]
- tests/NIT/nit.sh: fix portability for recursive "cp" [#3390]

## 2.53 2026-04-26 Jim Klimov <jimklimov+nut@gmail.com>

- m4/nut\_check\_libnetsnmp.m4: re-detect libnetsnmp details in mingw builds to not lose track [#3108]
- scripts/Windows/build-mingw-nut.sh: CI\_CACHE\_NUT\_HASHDIR\_CFG and configure\_nut(): stash a copy of first-run config.log and config.h as examples of resulting configuration [#3108]
- scripts/Windows/build-mingw-nut.sh: add © into the heading
- ci\_build.sh: get\_CI\_CACHE\_NUT\_HASHDIR\_CFG\_OPT(), configure\_nut(): stash a copy of first-run config.log and config.h as examples of resulting configuration [#3108]

## 2.54 2026-04-25 Jim Klimov <jimklimov+nut@gmail.com>

- docs/download.txt: fix NOTE block style
  - ci\_build.sh, Makefile.am: revise passing CI\_CACHE\_NUT\_HASHDIR\_CFG\_OPT into distcheck runs [#3108]
  - ci\_build.sh: consider\_cleanup\_shortcut(): do not fail if there is no out-of-tree CI\_BUILDDIR/Makefile.in - normally it should not be there (but do check for one in SCRIPTDIR)
  - ci\_build.sh: refactor get\_CI\_CACHE\_NUT\_HASHDIR\_CFG\_OPT() into a method, handle distcheck builds too [#3108]
  - NEWS.adoc: update about NIT caching of test certificates [#3390]
  - tests/NIT/README.adoc, docs/nut.dict: document caching [#3108, #3390, #1711]
-

- appveyor.yml, .circleci/config.yml: generalize to DO\_USE\_NUTCI\_CACHE, not just DO\_USE\_AUTOCONF\_CACHE [#3108, #3390]
- ci\_build.adoc, docs/nut.dict: document caching [#3108]
- ci\_build.sh, tests/NIT/nit.sh: generalize DO\_USE\_NUTCI\_CACHE so AUTOCONF and NIT\_TESTCERT toggles may be separate; drop old logic for config.cache in build area; rename CLEAN vars to ...\_BEFORE [#1711, #3390]
- tests/NIT/nit.sh: handle DO\_CLEAN\_AUTOCONF\_CACHE [#3390]
- tests/NIT/nit.sh: fix preparation of PEM and JKS client key/cert from OpenSSL [#1711, #3390] Move code that landed into a wrong case.
- tests/NIT/nit.sh: refactor preparation of PEM and JKS server key/cert from OpenSSL [#1711, #3390] Handle a situation with new openssl defaults and old(er) java keytool.
- tests/NIT/Makefile.am, tests/NIT/nit.sh: name the default WITH\_SSL\_TESTS=best-effort behavior for clarity [#1711]
- ci\_build.sh, scripts/Windows/build-mingw-nut.sh: ci\_build.sh: name CI\_CACHE\_NUT\_HASHDIR with AUTOCONF prefix [#3108]
- tests/NIT/nit.sh: prepare\_NIT\_certs(): rename TESTCERT\_VARS.sh to .env, generate one [#3390]

## 2.55 2026-04-24 Jim Klimov <jimklimov+nut@gmail.com>

- tests/NIT/nit.sh: fix NUT\_CAPATH for PERL SSL tests on Windows (MSYS2) [#1711]
- tests/NIT/nit.sh: prepare\_NIT\_certs(): for caching, also generate NSS DB from OpenSSL PEM if we can [#3390]
- tests/NIT/nit.sh: prepare\_NIT\_certs(): drop obsolete comment [#3390]
- tests/NIT/nit.sh: prepare\_NIT\_certs(): extend validity of generated test certificates to 20 years [#3390]
- tests/NIT/nit.sh: prepare\_NIT\_certs(): add generation of JKS (and more PEM) from NSS, but only if we plan to cache the result [#3390]
- tests/NIT/nit.sh: prepare\_NIT\_certs(): wrap long command lines [#3390]
- tests/NIT/nit.sh: refactor mock cert population logic as prepare\_NIT\_certs(), and allow caller-provided or automatically cached mocks [#3390]
- appveyor.yml: tweak notification triggers again [#1400]
- tests/NIT/nit.sh: fix NUT\_CAPATH for PERL SSL tests on Windows (MSYS2) [#1711]
- NEWS.adoc, clients/nutclient.cpp, clients/nutclient.h, clients/upsclient.c, conf/upsmon.conf.sample.in, docs/man/libnutclient\_tcp.txt, docs/man/upscli\_add\_host\_cert.txt: clients/{nut,ups}client, docs: make sure CERTHOST values for FORCESSL/CERTVERIFY are used (C++), allow inheriting global setting via -1 [#3331]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{cpp,h}: in TcpClient, track the applied SSLConfig\* vs. build(/run)time ability to use NSS (and/or) OpenSSL via UPSCLI\_SSL\_CAPS\* and consider that in getter preferences [#3331]
- clients/nutclient.cpp, clients/nutclient.h, docs/man/libnutclient\_commands.txt, docs/man/libnutclient\_tcp.txt: clients/nutclient.{cpp,h}, docs/man/libnutclient\_tcp.txt: add optional CERTHOST verification to OpenSSL code path [#3331]

## 2.56 2026-04-23 Jim Klimov <jimklimov+nut@gmail.com>

- clients/nutclient.cpp: some builds manage to get into WITH\_SSL\_CXX without WITH\_OPENSSL\NWITH\_NSS defined [#3331]
- m4/nut\_check\_libopenssl.m4, clients/upsclient.c, clients/nutclient.cpp, server/netssl.c: openssl\_cert\_verify\_san\_name(): support OpenSSL versions before ASN1\_STRING\_get0\_data() and potentially related helpers [#3331]
- ci\_build.sh: comment about the life-cycle of config.cache file [#3108]
- ci\_build.sh: align defaults for DO\_USE\_AUTOCONF\_CACHE and DO\_CLEAN\_AUTOCONF\_CACHE with existence of CI\_CACHE\_NUT\_BASEDIR [#3108]
- scripts/Windows/build-mingw-nut.sh: add support for CI\_TRACE envvar toggle, like in ci\_build.sh
- ci\_build.sh, scripts/Windows/build-mingw-nut.sh: pass CI\_CACHE\_NUT\_HASHDIR into Windows builds if we want to and can [#3108]
- NEWS.adoc, appveyor.yml: NEWS.adoc: ability to optionally re-use a config.cache file from older runs moved to NUT v2.8.6 [#3108]
- appveyor.yml, .circleci/config.yml: cache the out-of-tree locations for config.cache [#3108]
- ci\_build.sh: support reusable config.cache per "uname \+ script config opts" hash [#3108] Now even several build environments sharing a source directory with a generated configure script should be able to safely use their individual caches.
- ci\_build.sh: in developer mode (no BUILD\_TYPE) utilize make all-quick for less noise
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{cpp,h}: refactor SSLConfig\_CERTIDENT\_NSS to not construct with a separate certstore\_pass [#3331] NSS only has one pass phrase (shared for all private key database entries)
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{cpp,h}: refactor TcpClient and Socket to pass around SSLConfig(-derived) instances instead of private fields [#3331]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{cpp,h}: SSLConfig: add getCertHostBy\*() methods [#3331]
- clients/nutclient.cpp: clients/nutclient.{cpp,h}: SSLConfig\_\*::apply(): prepare and hide debug printouts [#3331]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{cpp,h}: SSLConfig\_CERTIDENT fix polymorphic-friendly reference to \_certstore [#3331]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclientmem.{cpp,h}: use \_empty\_str in constructors, just to be sure we have a valid reference to an immortal std::string where we expect one [#3331]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{cpp,h}: SSLConfig\*: introduce clone() methods to own data in our sets of CERTHOST/CERTSTORE/CERTIDENT and delete it properly [#3331]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{cpp,h}: SSLConfig: in the sets of CERTHOST/CERTSTORE/CERTIDENT data, retain polymorphic-friendly pointers instead of mutilated auto-casted objects [#3331]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{cpp,h}: SSLConfig::addCertHost(): constrain to adding non-trivial entries only [#3331]

## 2.57 2026-04-22 Jim Klimov <jimklimov+nut@gmail.com>

- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.{cpp,h}: move SSLConfig\* method implementations from header to CPP file [#3331]
  - clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.h: refactor SSLConfig\_NSS class to use new storage classes [#3331]
  - clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.h: refactor SSLConfig\_OpenSSL class to use new storage classes [#3331]
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- clients/nutclient.h: extend base SSLConfig class to keep track of optional CERTSTORE, CERTIDENT and/or CERTHOST info [#3331]
- clients/nutclient.cpp, clients/nutclient.h: clients/nutclient.h: introduce SSLConfig\* classes to keep CERTSTORE locations and credentials, and CERTIDENT and CERTHOST info [#3331]
- clients/nutclient.h: streamline code style for SSLConfig\* classes [#3331]
- clients/nutclient.cpp: Socket() constructor should not rely on openssl\_cert\_verify\_data struct to be pre-zeroed [#3331] OBS builds, and later valgrind, confirmed an issue with random decisions and free()'s based on random data in that struct. Apparently C differs from C++ in that aspect?..

## 2.58 2026-04-21 Jim Klimov <jimklimov+nut@gmail.com>

- drivers/usbhid-ups.c, NEWS.adoc: reconnect\_ups(): report success more visibly [#3423] Also fix setting "quiet" driver status when we actually remain in reconnecting state (or rather will soon re-enter it because situation is not fixed)
- clients/nutclient.cpp, clients/upsclient.c, server/netssl.c: openssl\_cert\_verify\_san\_name(): revise loop based on return from \*printf() [#3331]
- clients/nutclient.cpp, server/netssl.c: fix CI warnings [#3331]
- UPGRADING.adoc, clients/nutclient.cpp, clients/nutclient.h, docs/man/Makefile.am, docs/man/libnutclient\_tcp.txt, tests/NIT/nit.sh, tests/cpptest-client.cpp: clients/nutclient.{cpp,h}, tests/cpptest-client.cpp, nit.sh, docs: a CERTHOST address was missing (and subject/nickname is not the host name/url that NSS checks for) [#3331, #1711]
- clients/nutclient.cpp, clients/nutclient.h, docs/man/libnutclient\_tcp.txt: clients/nutclient.{cpp,h}: rename "forcessl" to match upsclient.c, and "tryssl" to not stand out [#3331]
- tests/cpptest-client.cpp: honour NUT\_DEBUG\_LEVEL by enabling (Socket Connect) debug toggle [#1711]
- tests/NIT/nit.sh: support passing individual NUT\_DEBUG\_LEVEL\_{PERL,PYTHON,CPPNIT,NUTSCAN} [#1711]
- server/netssl.c, clients/upsclient.c, clients/nutclient.cpp: align messages from NSS AuthCertificate() callbacks [#3331]
- clients/nutclient.cpp: Socket::startTLS(): if NSS handshake fails, add ssl\_url to the exception message [#3331]
- clients/nutclient.cpp: setSSLConfig\_\*(): debug-log the code path actually taken [#1711]
- clients/nutclient.cpp: Socket: document the SSL fields better [#3331]
- clients/nutclient.cpp: TcpClient::connect(): now we can config run-time for both NSS and OpenSSL, most options are shared and built-in abilities drive the final choice [#3331]
- tests/cpptest-client.cpp: setupClientSSL(): rearrange reporting of enabled SSL options [#1711]

## 2.59 2026-04-20 Jim Klimov <jimklimov+nut@gmail.com>

- tests/NIT/nit.sh: setenv\_ssl\_cppnit(): provide NUT\_CERTFILE and NUT\_KEYFILE values for OpenSSL tests [#1711, #3331]
  - clients/nutclient.cpp: transplant openssl\_cert\_verify\_callback() et al from upsclient.{c,h} [#3331]
  - clients/nutclient.cpp: Socket::startTLS(): some reported CN values (at least from openssl parser) may include escaped slashes as a separator [#3331]
  - clients/nutclient.cpp: fix syntax for NSS builds [#3331, #1711]
  - configure.ac: fix the checks for enabling SSL support in C++ builds [#1711, #3331]
  - server/netssl.c: ssl\_init(): some reported CN values (at least from openssl parser) may include escaped slashes as a separator [#3331]
  - server/netssl.c, server/nut\_ctype.h: transplant openssl\_cert\_verify\_callback() et al from upsclient.{c,h} [#3331]
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- clients/upsclient.c, clients/upsclient.h: clients/upsclient.{c,h}: revise code/comment markup for recent changes [#3331]
- tests/NIT/nit.sh: ensure SAN aliases (DNS and IP) that should NOT match the current host [#1711]
- clients/upsclient.c: upscli\_init2(): some reported CN values (at least from openssl parser) may include escaped slashes as a separator [#3331]
- clients/upsclient.c: openssl\_cert\_verify\_san\_name(): use X509\_check\_host()/X509\_check\_ip\_asc() where available [#3331]
- clients/upsclient.c: openssl\_cert\_verify\_callback(): consistently report depth along with CN (counterpart, CA...) [#3331]
- clients/upsclient.c: openssl\_cert\_verify\_callback(): do not call and log openssl\_cert\_verify\_san\_name() when already OK per caller [#3331]
- clients/upsched.c: hush down byte-by-byte read tracing (away from common debug level 6)
- clients/upsclient.c, clients/upsclient.h: clients/upsclient.c: convert openssl\_cert\_print\_san\_name() into openssl\_cert\_verify\_san\_name() [#3331] Also persist openssl\_cert\_verify\_data\_t in UPSCONN\_t so it is always available to callbacks when needed for this ups instance.
- clients/upsclient.c: openssl\_cert\_print\_san\_name(): handle IPv4/IPv6 addresses in SAN [#3331]
- clients/upsclient.c: rename example "mydata" into "openssl\_cert\_verify\_data" (types, vars) and example methods to print/verify certs [#3331]
- clients/upsclient.c, configure.ac: import openssl doc examples for certificate reporting/inspection into NUT styling and logging [#3331]
- clients/upsclient.c: fix markup about switch() so it is consistent in the file [#3331]
- clients/nutclient.cpp, clients/upsclient.c, server/netssl.c: server/netssl.c, clients/upsclient.c, clients/nutclient.cpp: when we verify subject\_CN[], consider comma as a separator too [#3331]
- clients/nutclient.cpp: fix a missed static\_cast [#3331]

## 2.60 2026-04-19 Jim Klimov <jimklimov+nut@gmail.com>

- clients/nutclient.cpp: be sure to include openssl/x509v3.h for methods we use [#3331]

## 2.61 2026-04-18 Tom Eldon <tom@tomeldon.com>

- drivers/apcmicrolink.c, drivers/apcmicrolink.h: drivers/apcmicrolink.c: minor handshake fix, allow flexible frame size Change to allow flexible frame size (upto 256 bytes) by waiting for page0, as some devices with payload size != 35 were incompatible. Allow variable length serial number string during handshake (as some devices use 14 characters).

## 2.62 2026-04-18 Jim Klimov <jimklimov+nut@gmail.com>

- drivers/apc\_modbus.c: addvar() the modbus\_retries option [#3414]

## 2.63 2026-04-16 Jim Klimov <jimklimov+nut@gmail.com>

- tests/NIT/nit.sh: setenv\_ssl\_common(): revise NUT\_CAPATH conditions [#1711]
  - tests/NIT/nit.sh: fix check of python FORCESSL is testable [#1711, #1349]
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## 2.64 2026-04-15 Jim Klimov <jimklimov+nut@gmail.com>

- scripts/python/module/PyNUT.py.in: fix sleep() [#1349, #1711]
- clients/nutclient.cpp, clients/nutclient.h, clients/upsclient.c, clients/upsclient.h, docs/man/libnutclient\_tcp.txt, docs/man/upscli\_ssl\_caps.txt, server/netssl.c, tests/NIT/nit.sh: Further differentiate reported CERTIDENT\_NAME/\_PASS capability [#1711]
- tests/NIT/nit.sh: refactor with setenv\_ssl\_common(), and in setenv\_ssl\_python() verify that we can run SSL tests (else do not force them) [#1349, #1711]
- scripts/python/module/PyNUT.py.in: detect inability to use SSL module (older interpreter etc.) [#1349, #1711]
- tests/NIT/nit.sh: align generation of CERTIDENT into upsd.conf/upsmon.conf with ability announced by the binaries [#1711]
- scripts/perl/test\_nutclient.pl: ANSI colors are a nice to have, do not crash if absent [#1711]
- scripts/perl/UPS/Nut.pm: StartTLS(): use a better portable "or" for "undef" printing [#1711]
- scripts/perl/test\_nutclient.pl: warnings are a nice to have, do not crash if absent [#1711]
- NEWS.adoc, clients/nutclient.cpp, clients/nutclient.h, clients/upsclient.c, clients/upsclient.h, docs/man/libnutclient\_tcp.txt, docs/man/upscli\_ssl\_caps.txt: clients/upsclient.c, clients/nutclient.cpp, headers, docs: propagate UPSCLI\_SSL\_CAPS\_CERTIDENT as a flag [#3331]
- server/netssl.c, clients/upsclient.c: \*ssl\_caps\_descr(): mention OpenSSL CERTIDENT (in)ability [#3331]
- scripts/perl/UPS/Nut.pm: warnings are a nice to have, do not crash if absent [#1711]
- clients/upsclient.c, clients/nutclient.cpp, server/netssl.c, m4/nut\_check\_libopenssl.m4: for OpenSSL password callback, check actual presence of methods, not version numbers [#3331]
- clients/upsclient.c: nss\_password\_callback(): reconcile markup with server/netssl.c [#3331]
- server/netssl.c: for now openssl\_password\_callback() should only be declared/implemented when OPENSSL\_VERSION\_NUMBER >= 1.1.0 [#3331]
- tests/NIT/nit.sh: account SKIPPED test cases (when isTestable\* says we can not) [#1711]
- tests/NIT/nit.sh: be portable, use TABCHAR instead of "\t" [#1711]
- docs/nut.dict: update for older systems
- configure.ac: work around ancient autotools without PACKAGE\_URL
- Makefile.am: hijack PARMAKES\_OPT to also set V=0 by default

## 2.65 2026-04-14 Lukas Schmid <lukas.schmid@netcube.li>

- drivers/apcmicrolink-maps.c, drivers/apcmicrolink-maps.h, drivers/apcmicrolink.c: drivers/apcmicrolink.c: try to use naming from apc\_modbus for some commands and values
- drivers/apcmicrolink.c: fix microlink\_page0\_state\_t init and reset

## 2.66 2026-04-14 Jim Klimov <jimklimov+nut@gmail.com>

- clients/upsclient.c: for now openssl\_password\_callback() should only be declared/implemented when OPENSSL\_VERSION\_NUMBER >= 1.1.0 [#3331]
  - conf/upsmon.conf.sample.in, docs/man/upsmon.conf.txt: clarify that CERTIDENT support requires OpenSSL 1.1.0\+ to fully work at the moment [#3331]
  - drivers/nutdrv\_qx.c, NEWS.adoc: qx\_is\_usb\_device\_supported(): only claim SUPPORTED if subdriver\_command was assigned [#3410]
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## 2.67 2026-04-14 Axel Gembe <axel@gembe.net>

- NEWS.adoc, drivers/apc\_modbus.c: apc\_modbus: increase default TCP response timeout to 2000 ms The default 500ms timeout value is marginal for Modbus TCP, even on a fast local network. In tests with an APC UPS, we see timeouts because of this: `` Apr 14 19:10:44 ares apc\_modbus[2002143]: \_apc\_modbus\_read\_registers: Read of 0:27 failed: Connection timed out (192.168.0.102:502) Apr 14 19:10:44 ares apc\_modbus[2002143]: \_apc\_modbus\_read\_registers: Read of 0:27 failed: Invalid data (192.168.0.102:502) Apr 14 19:10:44 ares apc\_modbus[2002143]: \_apc\_modbus\_close: Closing connection Apr 14 19:10:47 ares apc\_modbus[2002143]: Opened modbus successfully `` Packet capture: `` No. Time Source Destination Protocol Length Info 3636 2026-04-14 19:10:43.842072 192.168.0.40 192.168.0.102 Modbus/TCP 66 Query: Trans: 12622; Unit: 1, Func: 3: Read Holding Registers 3637 2026-04-14 19:10:44.018369 192.168.0.102 192.168.0.40 TCP 60 502 → 40392 [ACK] Seq=70441 Ack=11905 Win=3660 Len=0 3638 2026-04-14 19:10:44.377723 192.168.0.40 192.168.0.102 Modbus/TCP 66 Query: Trans: 12623; Unit: 1, Func: 3: Read Holding Registers 3639 2026-04-14 19:10:44.578103 192.168.0.102 192.168.0.40 TCP 60 502 → 40392 [ACK] Seq=70441 Ack=11917 Win=3648 Len=0 3640 2026-04-14 19:10:44.579449 192.168.0.102 192.168.0.40 Modbus/TCP 117 Response: Trans: 12622; Unit: 1, Func: 3: Read Holding Registers 3641 2026-04-14 19:10:44.579482 192.168.0.40 192.168.0.102 TCP 54 40392 → 502 [ACK] Seq=11917 Ack=70504 Win=63 Len=0 3642 2026-04-14 19:10:44.579604 192.168.0.40 192.168.0.102 TCP 54 40392 → 502 [FIN, ACK] Seq=11917 Ack=70504 Win=63 Len=0 `` 3636 is the first read, 3638 is the retry after 500ms without response, both are ACKed. 3640 is the actual response to transaction 12622, which arrives 700ms after the initial packet. At this point we expect a response to transaction 12623, which is why we fail with "invalid data". To fix this we increase the default TCP response timeout to 2000 ms, which seems to be more reasonable for a device that already can take 700 ms to respond under ideal conditions. I chose a bit higher value for users who might connect the device over a higher latency network.
- NEWS.adoc, docs/nut.dict, drivers/apc\_modbus.c, drivers/apc\_modbus.h: apc\_modbus: decode RunTimeCalibrationStatus\_BF RunTimeCalibrationStatus\_BF is explained in MPAO-98KJ7F\_R1\_EN Appendix B. This adds defines for the bitfield and decodes it into `experimental.ups.calibration.result` using a string join converter.
- NEWS.adoc, docs/nut.dict, drivers/apc\_modbus.c, drivers/apc\_modbus.h: apc\_modbus: decode PowerSystemError\_BF as alarms PowerSystemError\_BF is explained in MPAO-98KJ7F\_R1\_EN Appendix B. This adds defines for the bitfield and decodes it as alarms in the driver.
- NEWS.adoc, docs/nut.dict, drivers/apc\_modbus.c, drivers/apc\_modbus.h: apc\_modbus: decode BatterySystemError\_BF as alarms BatterySystemError\_BF is explained in MPAO-98KJ7F\_R1\_EN Appendix B. We were already using the bitfield for the RB status, this adds defines for the bitfield and decodes it as alarms in the driver.
- NEWS.adoc, docs/man/apc\_modbus.txt, drivers/apc\_modbus.c: apc\_modbus: simplify error handling with read retries Replace the complex `_apc_modbus_handle_error` function with a simpler retry mechanism built into `_apc_modbus_read_registers`. The new approach: - Retries register reads on ETIMEDOUT errors up to `modbus_retries` times (configurable, default 3) - On non-timeout errors or after retry exhaustion, closes the connection for reconnection on the next update cycle - Removes the platform-specific (WIN32/POSIX) timeout detection and the flush-based recovery that didn't work anyway (flush is already done in `_apc_modbus_reopen` upon reconnection) Also adds a `modbus_retries` driver option to configure the number of read retry attempts, and improves logging for connection open/close events. This change was inspired by a patch by @marcan to do the same and from testing the behaviour of `apcupsd`, noticing that on my USB unit it times out on read and only succeeds on the first retry.
- NEWS.adoc, drivers/apc\_modbus.c: apc\_modbus: always zero terminate string join `_apc_modbus_string_join` did not zero terminate the result in every case, like when all values are NULL.
- drivers/apc\_modbus.c: apc\_modbus: bump driver version to 0.20

## 2.68 2026-04-13 Lukas Schmid <lukas.schmid@netcube.li>

- docs/nut.dict: Add more words to dict
- drivers/apcmicrolink.c: fix most static compile issues
- docs/nut.dict: add apcmicrolink and RMI

## 2.69 2026-04-13 Jim Klimov <jimklimov+nut@gmail.com>

- NEWS.adoc: introduce apcmicrolink driver [#3406]
  - drivers/Makefile.am: EXTRA\_DIST apcmicrolink-maps.h apcmicrolink.h [#3406]
  - docs/man/Makefile.am: provide apsmicrolink.html also [#3406]
  - tests/NIT/nit.sh: generate upsmon.pem [#3331, #1711]
  - NEWS.adoc, clients/nutclient.cpp, clients/upsclient.c, conf/upsd.conf.sample, conf/upsmon.conf.sample.in, docs/man/upsd.conf.txt, docs/man/upsmon.conf.txt, server/netssl.c: upsd, libupsclient, libnutclient, upsmon: in OpenSSL builds, support CERTPATH pointing to a big PEM file as well as a directory [#3331]
  - clients/nutclient.cpp, clients/nutclient.h, tests/cpptest-client.cpp: clients/nutclient.{h,cpp}, tests/cpptest-client.cpp: extend setSSLConfig\_OpenSSL() etc. to support CERTIDENT name (validation of client' own cert) [#3331]
  - clients/upsclient.c: fix CERTIDENT (OpenSSL) to check for CN=... part too [#3331]
  - server/netssl.c: fix CERTIDENT (OpenSSL) to check for CN=... part too [#3331]
  - tests/NIT/nit.sh: generate upsmon.pem [#3331, #1711]
  - tests/NIT/nit.sh: setenv\_ssl\_cpptest(): fix check for TESTCERT\_PATH\_ROOTCA [#1711]
  - tests/NIT/nit.sh: unlock OpenSSL features that are now on par with NSS [#3331, #1711]
  - scripts/python/module/PyNUT.py.in, scripts/perl/UPS/Nut.pm: handle CERTIDENT to validate loaded client certificate name [#3331] Make sure the file we got is the file we wanted. Belts and suspenders rule!
  - scripts/python/module/PyNUT.py.in, scripts/perl/UPS/Nut.pm: handle CERTHOST (dict or list of dict/tuple) to validate connections better [#3331]
  - scripts/perl/UPS/Nut.pm: StartTLS(): alias variables named like in other NUT code into SSL\_\* if not provided directly [#1711]
  - clients/nutclient.cpp, clients/upsclient.c, conf/upsd.conf.sample, docs/man/upsd.conf.txt, server/netssl.c: clients/nutclient.cpp, clients/upsclient.c, server/netssl.c: reconcile variants of openssl\_password\_callback() [#3331] Notably, in C code (upsclient, netssl) constrain with OpenSSL >= 1.1 support, and do use SSL\_CTX\_set\_default\_passwd\_cb\_userdata() to pass the configured (or generally otherwise provided) certpasswd string.
  - NEWS.adoc, clients/upsmon.c, conf/upsmon.conf.sample.in, docs/man/upsmon.conf.txt: clients/upsmon.c, NEWS.adoc, conf/, **docs/man/**: introduce support of CERTFILE to upsmon [#3331]
  - clients/upsclient.c, clients/upsclient.h, docs/man/Makefile.am, docs/man/upscli\_init.txt: clients/upsclient.{c,h}, docs/man/upscli\_init.txt: implement client CERTIDENT correctly (via CERTFILE); introduce upscli\_init2() [#3331]
  - NEWS.adoc, conf/upsd.conf.sample, docs/man/upsd.conf.txt, server/conf.c, server/netssl.c, server/netssl.h: server/\*, NEWS.adoc, conf/upsd.conf.sample, docs/man/upsd.conf.txt: implement server-side CERTPATH for OpenSSL [#3331] Optionally use for validation of clients with OpenSSL collection of CA PEM files.
  - conf/upsd.conf.sample, docs/man/upsd.conf.txt, server/conf.c, server/netssl.c: server/\*, conf/upsd.conf.sample, docs/man/upsd.conf.txt: implement server-side CERTIDENT for OpenSSL [#3331]
  - NEWS.adoc, conf/upsd.conf.sample, conf/upsmon.conf.sample.in, docs/man/upsd.conf.txt, docs/man/upsmon.conf.txt: NEWS.adoc, docs/man/, **conf/**: update docs about newly supported features with OpenSSL [#3331]
  - server/conf.c: reshuffle options processing and update comments now that CERTREQUEST is for both backends and more may follow [#3331]
  - clients/upsclient.c: upscli\_sslinit(): consider CERTHOST→certverify setting to NOT abort the connection if verification was not required (0) [#3331]
  - clients/upsclient.c: upscli\_sslinit(): consider OPENSSL\_VERSION\_NUMBER constraint in logging, possibly rejecting the StartTLS attempt with OpenSSL too old (can not fulfil required security, avoid sad mishaps) [#3331]
-

- `conf/upsd.conf.sample`, `docs/man/upsd.conf.txt`: heads-up about MAXCONN and ulimit [#3365]
- NEWS.adoc: remove duplicated paragraph (silent merge conflict)
- `clients/upsmon.c`, `clients/upsrw.c`, `common/common.c`, `drivers/adelsystem_cbi.c`, `drivers/generic_modbus.c`, `drivers/main.c`, `drivers/powerman-pdu.c`, `drivers/snmp-ups.c`, `drivers/tripplite_usb.c`, `tools/nut-scanner/nutscan-serial.c`: **.c: fix use of non-const char** with `strstr()` and `strchr()` return values [#823]
- `docs/man/Makefile.am`: fix possible non-delivery of NUTSCAN\_UPSLOG\_SET\_DEBUG\_LEVEL\_DEPS pages [#3378, #3379]
- `docs/man/upscli_init.txt`: update wording and markup for pre-existing features [#3331]
- NEWS.adoc: announce the adjusted markup format

## 2.70 2026-04-12 Jim Klimov <jimklimov@gmail.com>

- `scripts/perl/UPS/Nut.pm`: warn if CERTVERIFY and custom CA options are enabled on Darwin platform [#3404]
- `scripts/perl/UPS/Nut.pm`: align closer with `can_ssl` [#1711]
- `scripts/perl/UPS/Nut.pm`, `scripts/perl/test_nutclient.pl`, `tests/NIT/nit.sh`: `scripts/perl/*`, `tests/NIT/nit.sh`: streamline passing NUT\_DEBUG (or defaulting to no debug) [#1711]
- `tests/NIT/nit.sh`: neuter CERTVERIFY on darwin for now [#3404]
- `tests/NIT/nit.sh`: centralize definition of PERL\_OPTS [#1711]
- `scripts/perl/UPS/Nut.pm`: allow SSL cert validation for IP addresses as host names [#1711]
- `scripts/perl/UPS/Nut.pm`: pass only defined args to `IO::Socket::SSL→start_SSL()` [#1711]
- `scripts/perl/UPS/Nut.pm`: support debugging messages of `IO::Socket::SSL` [#1711]
- `tests/NIT/nit.sh`: report the non-default NIT\_CASE being handled [#1711]

## 2.71 2026-04-12 Jim Klimov <jimklimov+nut@gmail.com>

- `clients/upsclient.c`: `upscli_sslinit()`: OPENSSL\_VERSION\_NUMBER constraint applies to whole block for X509\_VERIFY\_PARAM [#3331]
  - `server/netssl.c`: extend OpenSSL builds feature set to support CERTREQUEST validation of clients [#3331]
  - `clients/upsclient.c`: extend OpenSSL builds feature set to support pinning server certificates [#3331]
  - `clients/upsclient.c`: extend OpenSSL builds feature set to support a client certificate\+key, optionally with password [#3331]
  - `scripts/perl/UPS/Nut.pm`, `scripts/perl/test_nutclient.pl`: `scripts/perl/UPS/Nut.pm`: now we can pass SSL debug level as a number [#1711]
  - `scripts/perl/UPS/Nut.pm`: fix to loading `IO::Socket::SSL` via `eval` with quotes not braces [#1711] It is then evaluated at run-time as code reaches that line (or not - conditionals), not at interpretation time where it fails too early. Also use `"1;"` in the end as a tidy practice.
  - `clients/nutclient.cpp`: work around lack of threads in some mingw versions [#3402]
-

## 2.72 2026-04-11 Lukas Schmid <lukas.schmid@netcube.li>

- data/driver.list.in, docs/man/Makefile.am, docs/man/apcmicrolink.txt, drivers/Makefile.am, drivers/apcmicrolink-maps.c, drivers/apcmicrolink-maps.h, drivers/apcmicrolink.c, drivers/apcmicrolink.h: drivers: add experimental APC Microlink serial driver Implement a new apcmicrolink driver for APC Smart-UPS units that expose the Microlink serial protocol. The driver establishes a Microlink session over the serial link, reads the device descriptor blob, and uses that runtime descriptor map to publish standard NUT variables, writable settings, status bits, alarms, and supported instant commands. Add labeled Microlink command masks for battery tests, calibration, UPS control, and outlet-group power control; wire APC-style load and shutdown commands through the static command map; publish input.transfer.reason from UPS status change cause; and stop folding simple-signaling status bits into ups.status. Also hook upsdrv\_shutdown() up to the Microlink shutdown request, document the driver, and clean up initial upstream review issues such as missing file headers, whitespace, and format-string warnings.

## 2.73 2026-04-11 Jim Klimov <jimklimov+nut@gmail.com>

- m4/ax\_c\_pragmas.m4, clients/nutclient.cpp: introduce HAVE\_PRAGMA\_GCC\_DIAGNOSTIC\_IGNORED\_DEPRECATED\_DECLARATIONS to implement methods for deprecated "master" operation [#840, #3402]

## 2.74 2026-04-10 Jim Klimov <jimklimov+nut@gmail.com>

- NEWS.adoc: announce updates to client binding libraries [#3402]
- scripts/perl/test\_nutclient.pl: avoid "Can't call method "Error" on an undefined value" messages in failed tests; clearly say that \$nut is no more [#1711]
- scripts/perl/UPS/Nut.pm: when IO::Socket::SSL is not available, report the error with "FEATURE-NOT-SUPPORTED" in text [#1711]
- tests/NIT/nit.sh: setenv\_ssl\_perl(): check if IO::Socket::SSL is available, neuter NUT\_FORCESSL based on that [#1711]
- scripts/perl/test\_nutclient.pl: support USESSL=undef if no value provided in env [#1711]
- scripts/perl/UPS/Nut.pm: support USESSL=undef, check that we can\_ssl and follow up with that [#1711]
- scripts/perl/UPS/Nut.pm: \_initialize(): include previously reported error in SSL setup fail messages [#1711]
- scripts/perl/UPS/Nut.pm: apply indentation consistently (3 styles were intermixed)
- scripts/perl/UPS/Nut.pm, scripts/perl/test\_nutclient.pl: support NUT\_SSL envvar toggle to disable attempts at SSL altogether [#1711]
- scripts/perl/UPS/Nut.pm, scripts/perl/test\_nutclient.pl: SSL\_ca\_path should be defined in args by caller, not by a hack in production code [#1711]
- scripts/perl/UPS/Nut.pm: STARTTLS: debug-print %arg list that is passed to the SSL library [#1711]
- scripts/perl/UPS/Nut.pm: use CAPATH to provide SSL\_ca\_path, not SSL\_ca\_file [#1348, #1711]
- scripts/python/module/PyNUT.py.in: retry SSL handshake if it timed out, a few times [#3401]
- server/netssl.c: net\_starttls(): report which SSL backend is in place (if any) [#3331]
- scripts/python/module/PyNUT.py.in: isValidProtocolVersion(): stringify and strip the "result" before matching in regex [#1349]
- clients/Makefile.am: bump libupsclient ABI version
- tests/NIT/nit.sh: add a note about jNut using this script [#1711, #1350]
- clients/nutclient.cpp, clients/nutclient.h, clients/nutclientmem.cpp, clients/nutclientmem.h, tests/cpptest-client.cpp: clients/nutclient{,mem}.{h,cpp}, tests/cpptest-client.cpp: remember if we enabled TRACKING so it is really done once, confirm with isTrackingModeEnabled() [#656]

- scripts/perl/test\_nutclient.pl: fix DeviceLogin⇒ListClient for "admin" without a role [#1711]
- scripts/perl/UPS/Nut.pm: \_send(): reset the error buffer before networking away [#1711, #1348]
- scripts/perl/UPS/Nut.pm: we can only authenticate once per connection, skip other attempts [#1711, #1348]
- docs/download.txt: publish jNut-1.1 [#1350]
- NEWS.adoc: announce updates to client binding libraries [#3402]
- appveyor.yml: post only one (custom) notification per build [#1400]
- docs/man/Makefile.am: hush "cat" errors in LIBNUTCLIENT\_TCP\_DEPS\_SUB recipe

## 2.75 2026-04-09 Jim Klimov <jimklimov+nut@gmail.com>

- scripts/perl/test\_nutclient.pl: fix reporting of @failed methods [#1711]
  - scripts/perl/UPS/Nut.pm: GetTrackingResult(): fix parsing results of GET TRACKING [#1348]
  - tests/NIT/nit.sh: when we ask for specific NIT\_CASE values (cppnit, python, perl, nutscanner) the lack of prerequisites should be a FAILURE [#1711]
  - tests/NIT/nit.sh: introduce tests for PERL [#1348, #1711]
  - message, so we can parse it [#1348, #1711]
  - scripts/perl/test\_nutclient.pl: fix access to possibly undefined envvars [#1348, #1711]
  - scripts/perl/UPS/Nut.pm: fix error messages when Authenticate() and Login() fail [#1348, #1711]
  - scripts/perl/UPS/Nut.pm, scripts/perl/test\_nutclient.pl: scripts/perl/test\_nutclient.pl: fix test case setup, update comments [#1711, #1348]
  - scripts/perl/UPS/Nut.pm: fix string work with TRACKING logic
  - scripts/perl/UPS/Nut.pm: change commands that address an UPS to optionally accept the UPS name as argument, only using
  - scripts/python/module/test\_nutclient.py.in: fix typo in debug traces
  - assignments into debug() calls [#1348, #1711]
  - COPYING, scripts/HP-UX/nut.psf.in, scripts/Makefile.am, scripts/obs/debian.libups-nut-perl.install, scripts/perl/{ ⇒ UPS}/Nut.pm, scripts/perl/test\_nutclient.pl: scripts/perl/test\_nutclient.pl: introduce tests, move Nut.pm to UPS subdir [#1348, #1711]
  - scripts/perl/Nut.pm: call StartTLS() from \_initialize(), and use isValidProtocolVersion() method to actively ping a connection after STARTTLS claimed success (but handshake might be broken in fact) [#3387, #1348]
  - clients/nutclient.cpp, clients/upsclient.c, scripts/python/module/PyNUT.py.in: Client libraries (C, C+++, Python): use isValidProtocolVersion() method to actively ping a connection after STARTTLS claimed success (but handshake might be broken in fact) [#3387]
  - clients/nutclient.cpp, clients/nutclient.h, clients/nutclientmem.cpp, clients/nutclientmem.h, clients/upsclient.c, clients/upsclient.h, scripts/perl/Nut.pm, scripts/python/module/PyNUT.py.in: Client libraries (C, C+++, Python, PERL): introduce an isValidProtocolVersion() method to actively ping a connection [#3387]
  - clients/nutclient.cpp, clients/nutclient.h, clients/nutclientmem.cpp, clients/nutclientmem.h: clients/nutclient{,mem}.{h,cpp}: becomeSecondary() was overkill, there is no such verb in NUT protocol (LOGIN already reaches that role level)
  - scripts/python/module/PyNUT.py.in: refactor FSD() with an explicit becomePrimary() method
  - scripts/perl/Nut.pm: becomeSecondary() was overkill, there is no such verb in NUT protocol (LOGIN already reaches that role level)
  - scripts/perl/Nut.pm: add warnings toggle
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- scripts/perl/Nut.pm: fix typos in comments
- scripts/perl/Nut.pm: rename primary() $\Rightarrow$ becomePrimary, add becomeSecondary() for completeness [#1348]
- clients/nutclient.cpp, clients/nutclient.h, clients/nutclientmem.cpp, clients/nutclientmem.h: clients/nutclient{,mem}.{h,cpp}: rename primary() $\Rightarrow$ becomePrimary, add becomeSecondary() for completeness [#656]
- tests/cpptest-client.cpp: add test with SET VAR TRACKING [#1711, #656]
- scripts/python/module/test\_nutclient.py.in: add test with SET VAR TRACKING [#1711, #1349]
- scripts/perl/Nut.pm: fix typos in comments
- docs/download.txt: refer to new release of jNut-1.0 (and the following snapshot evolution)
- tests/NIT/nit.sh: do not unset DEBUG\_SLEEP when we generate configs [#1711] We might actually want more of those, with test case passwords revealed, etc.

## 2.76 2026-04-08 Jim Klimov <jimklimov+nut@gmail.com>

- scripts/python/module/PyNUT.py.in: TrackingID class: include timestamp tracking and age reporting [#1349]
  - scripts/perl/Nut.pm: TrackingID class: include timestamp tracking and age reporting [#1348]
  - clients/nutclient.cpp, clients/nutclient.h, clients/nutclientmem.cpp, clients/nutclientmem.h: clients/nutclient{,mem}.{h,cpp}, tests/\*.cpp: implement waiting for TRACKING result if requested [#656]
  - clients/nutclient.h, clients/Makefile.am: Implement TrackingID as a separate class, including timestamp tracking and age reporting [#656]
  - scripts/python/module/PyNUT.py.in: Implement TrackingID as a separate class [#1349]
  - scripts/python/module/PyNUT.py.in: fix syntax [#1349]
  - scripts/perl/Nut.pm: Implement TrackingID as a separate class [#1348]
  - scripts/perl/Nut.pm: fix syntax [#1348]
  - scripts/python/module/PyNUT.py.in: revise AI creations, complete the TRACKING\+waiting dialog support [#1349]
  - scripts/perl/Nut.pm: revise AI creations, complete the TRACKING\+waiting dialog support [#1348]
  - scripts/perl/Nut.pm, scripts/python/module/PyNUT.py.in: AI PoC: implement the remainder of NUT Networked protocol (as of NUT v2.8.5) for Python and Perl bindings
  - scripts/perl/Nut.pm: clean up trailing white-space
  - tests/NIT/nit.sh: simplify with NIT\_CASE=generatecfg\_sandbox support [#1711]
  - tests/NIT/nit.sh: allow injecting WITH\_SSL\_CLIENT, WITH\_SSL\_SERVER, WITH\_SSL\_SERVER\_CLIVAL (e.g. to test against older packaged builds) [#1711]
  - .github/workflows/05-codeql.yml: drop duplicate init stage
  - configure.ac: hush clang-21 "-Wc11-extensions" warnings about system headers
  - docs/download.txt: fix NOTE markup for N4W 7Zip size
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**2.77 2026-04-07 Jim Klimov <jimklimov+nut@gmail.com>**

- docs/maintainer-guide.txt: revise instructions for post-release steps (website, etc.)
- docs/download.txt: newly refer to nut-X.Y.Z-docs.tar.gz archives
- docs/docinfo.xml.in: fix release date of v2.8.5 and move its tag to be above v2.8.4
- NEWS.adoc, UPGRADING.adoc, docs/docinfo.xml.in: Revert "NEWS.adoc, UPGRADING.adoc, docs/docinfo.xml.in: finalize text before NUT v2.8.5 release" This reverts commit 784219a3dc935e53a1273063c591bc6dd949590b: finally released NUT v2.8.5, preparing for v2.8.6 cycle now.

**2.78 2026-04-06 Jim Klimov <jimklimov+nut@gmail.com>**

- common/wincompat.c: syslog(), send\_to\_named\_pipe(): debug-log when we failed to open existing NAMED\_PIPE and so move on [#3302]
- common/wincompat.c: log errors after methods faults via upslog\_with\_errno() not plain upslogx() [#3302]
- common/wincompat.c: log progress through pipe\*() methods [#3302, #3368]
- common/wincompat.c: fix send\_to\_named\_pipe() to measure strlen(data) once [#3302]
- common/wincompat.c: fix getpass() to measure strlen(wincompat\_password) once [#3302]
- common/wincompat.c: fix filter\_path() to measure strlen(source) once [#3302]
- common/wincompat.c: tcflush(): fix bitwise OR to boolean OR

**2.79 2026-03-31 Jim Klimov <jimklimov+nut@gmail.com>**

- server/upsd.c: sendback(): adjust debug traces for write() vs ssl\_write() [#3379, #3331]
- server/netssl.c: net\_starttls(): send\_err\_extra(NUT\_ERR\_ACCESS\_DENIED) for faults in NSS setup [#3379, #3331]
- server/upsd.c, server/upsd.h: server/upsd.{c,h}: introduce send\_err\_extra() [#3331]
- drivers/dstate.c, NEWS.adoc: revise closing of connections and returning error codes; document the methods [#3302]

**2.80 2026-03-30 Jim Klimov <jimklimov+nut@gmail.com>**

- drivers/dstate.c: debug-log the progress around starting and maybe failing CreateNamedPipe() [#3302]
- drivers/dstate.c: sock\_disconnect(): resync code with wincompat::pipe\_disconnect() implementation [#3302]
- drivers/dstate.c: sock\_disconnect(): try to flush the buffer [#3368]

**2.81 2026-03-25 Jim Klimov <jimklimov+nut@gmail.com>**

- drivers/dstate.c: err on the safe side and set conn=NULL after sock\_disconnect() [#3368]

**2.82 2026-03-24 Jim Klimov <jimklimov+nut@gmail.com>**

- drivers/upsdrvctl.c: refactor with forkexec\_parent\_analyze() and a re-check before retry - also for WIN32 [#3302] Also revised WaitForSingleObject() result checking - there has to be a chance to succeed ;)
  - drivers/upsdrvctl.c: refactor with forkexec\_parent\_analyze() and a re-check before retry - POSIX part [#3302]
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### 2.83 2025-10-02 Jim Klimov <jimklimov+nut@gmail.com>

- configure.ac: WIN32 GetAdaptersInfo(): be careful about \$NETLIBS\_GETADDRS usage
  - ci\_build.sh: retain config.cache if DO\_CLEAN\_AUTOCONF\_CACHE=no also in dev-test builds [#3108]
  - ci\_build.sh: configure\_nut(): log if config.cache is available and if we want/do use it [#3108]
  - ci\_build.sh: do not default to DO\_USE\_AUTOCONF\_CACHE=yes/DO\_CLEAN\_AUTOCONF\_CACHE=no [#3108] Callers like specific CI recipes should be free to enable certain behaviors though.
  - .circleci/config.yml: default to DO\_CLEAN\_AUTOCONF\_CACHE:no DO\_USE\_AUTOCONF\_CACHE:yes and carry over the config.cache between builds [#3108]
  - appveyor.yml: default to DO\_CLEAN\_AUTOCONF\_CACHE:no DO\_USE\_AUTOCONF\_CACHE:yes and carry over the config.cache between builds [#3108]
  - ci\_build.sh, scripts/Windows/build-mingw-nut.sh, NEWS.adoc: introduce options to USE\_AUTOCONF\_CACHE and/or CLEAN\_AUTOCONF\_CACHE [#3108]
-